

MBA 753: Causal Inference Models - Assignment 1

240614-Manan

Q1. Using UNHDD data, fit a regression model for GNI (Gross National Income) with the partial effects of HDI (Human Development Index), GII (Gender Inequality Index), life expectancy at birth, and population age 15-64 years (in millions)

a) Which effects are significant?

Looking at the output, HDI ($p < 0.001$) and GII ($p = 0.002$) are the only significant predictors at the 5% level. Both life expectancy and the working-age population (15-64 years) do not have a statistically significant impact in this model, as their p-values are well above 0.05.

b) Perform model diagnostics. Mention the assumptions tested, methods used and their corresponding results.

Testing the standard OLS assumptions based on the provided plots and VIF scores:

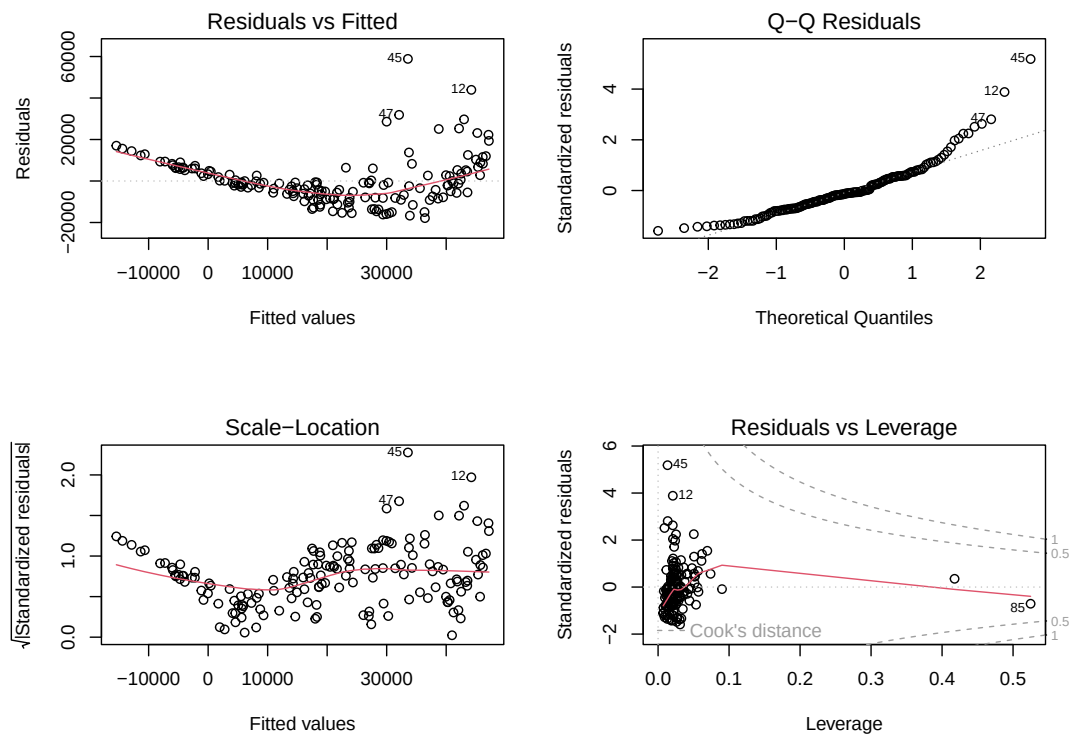


Figure 1: Model diagnostics for the base regression model (Q1).

- **Linearity:** The Residuals vs Fitted plot shows a mostly flat red line, meaning it is fair to assume linearity.

- **Normality:** In the Q-Q plot, the residuals hug the line nicely in the middle but drift off significantly at the top right (around observations 12, 45, and 47). This right-skewness shows that our normality assumption is violated.
- **Homoscedasticity:** The Scale-Location plot reveals a pretty obvious funnel shape. Since the points fan out as the fitted values increase, we definitely have heteroscedasticity (non-constant variance).
- **Influential Observations:** The Residuals vs Leverage plot shows most data points are safe, but a couple of outliers (like point 45) have higher leverage and might be pulling the regression coefficients off balance.
- **Multicollinearity:** The VIF for HDI is 10.66. Because this is over the usual threshold of 10, there's a severe multicollinearity issue between HDI and the other variables in the model.

c) Interpret the regression coefficients.

Keeping all other variables constant, we can interpret the coefficients as follows:

- **Intercept (\$4,862.53):** This is the average GNI if all other predictors were completely zero. It doesn't mean much in practice because a country can't literally have a zero for HDI or life expectancy.
- **HDI (\$85,517.64):** For every 1-unit jump in the Human Development Index, on an average a country's GNI is expected to go up by roughly \$85,517.
- **GII (-\$37,769.14):** A 1-unit increase in the Gender Inequality Index drops the average GNI by about \$37,769.
- **Life Expectancy & Pop 15-64:** Since these weren't statistically significant, we can't really make any solid claims about their effect on GNI using this specific model.

Q2. In the above regression model, replace HDI with HDI groups (all other predictors have to be retained in the model). Use "low" as the reference category for HDI groups.

a) How many categories of HDI group are present in the dataset?

There are 4 HDI groups in the dataset: Low, Medium, High, and Very high.

b) Interpret the regression coefficients of HDI groups? Are they in tune with the results from question 1?

Using "Low" as our baseline category:

- **Medium and High:** Neither of these groups shows a statistically significant difference from the Low group ($p > 0.30$).
- **Very high (\$15,578.20):** This group is highly significant ($p = 0.003$). It tells us that countries in the "Very high" category have an expected GNI that is \$15,578 higher than those in the "Low" category, assuming everything else is constant.

Yes, this lines up perfectly with what we saw in Q1. The continuous HDI variable showed a strong positive link with GNI, and this categorical breakdown confirms that the top tier of development is associated with a massive jump in wealth compared to the bottom tier.

c) Change the reference group to “Very High” and fit the model. Interpret the regression coefficients of the categorical predictor. What are your conclusions about the model?

Switching the baseline to “Very High” makes all the other categories show up with significant, negative coefficients:

- **Low:** Expected GNI is \$15,578.20 lower than the Very High group.
- **Medium:** Expected GNI is \$18,884.80 lower than the Very High group.
- **High:** Expected GNI is \$19,024.90 lower than the Very High group.

Conclusion: The “Very high” HDI group is in a financial league of its own. Even though the Low, Medium, and High groups don’t differ much from one another (as we saw in 2b), they are all significantly poorer than the Very High category.

Q3. In the regression model from question 2, include an interaction term between HDI groups and GII.

a) Is the interaction significant?

Yes, it is highly significant. The ANOVA F-test gives us a p-value of roughly 4.98×10^{-5} , which is way below the 0.05 cutoff. This means the interaction between the HDI group and GII definitely belongs in the model.

b) Interpret the interaction effects.

The interaction shows that gender inequality impacts a country’s GNI differently depending on how developed the country is:

- For our baseline (“Low” HDI), the effect of GII isn’t even statistically significant ($p = 0.771$). So, gender inequality doesn’t measurably hurt GNI in low-development countries.
- However, for “Very high” HDI countries, the interaction term is heavily negative (-89872.6) and very significant ($p = 0.001$). This means the negative impact of gender inequality on GNI is vastly worse for highly developed nations than it is for low-development ones.

c) Perform model diagnostics and report their results.

- **Multicollinearity:** The GVIF output flags extremely high values (e.g., nearly 6000 for the interaction term). However, this kind of structural multicollinearity is completely normal and mathematically unavoidable when you multiply variables together to create an interaction term.
- **Visual Diagnostics:** Based on the visual constraints from the model, we still have issues with non-normality at the upper tail and heteroscedasticity. This might bias standard errors a bit, but the interaction is so overwhelmingly significant that the core finding remains solid.

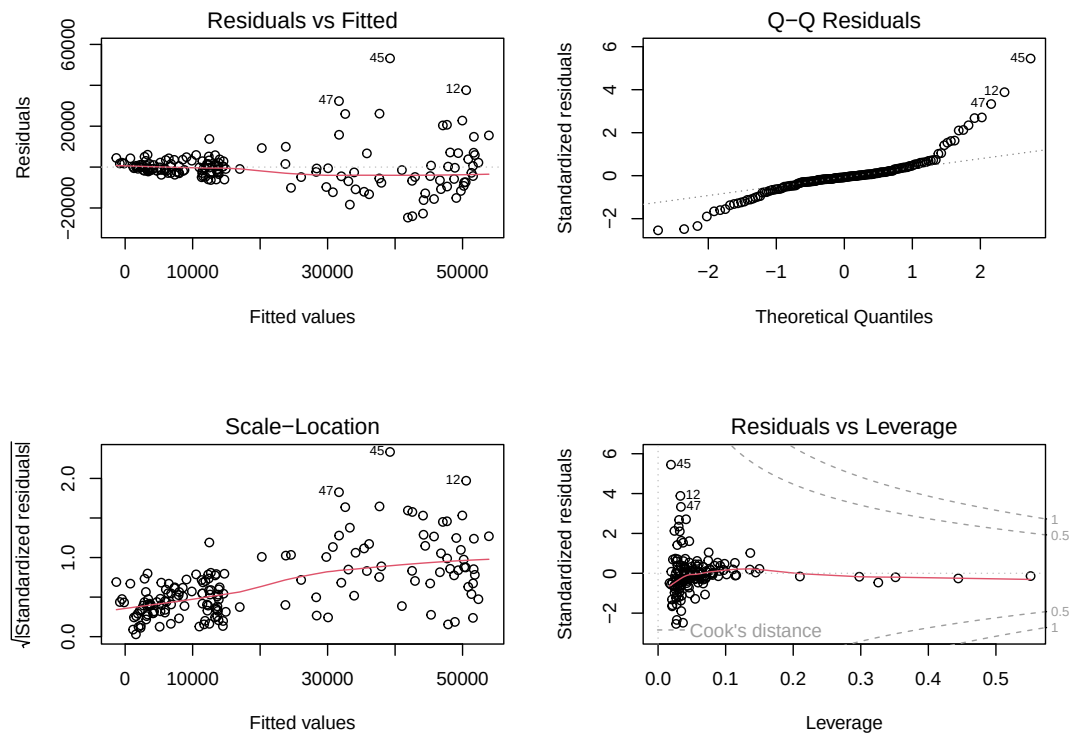


Figure 2: Model diagnostics for the base regression model (Q3).

d) What are your conclusions about the regression model?

Adding the interaction term gives us a much better, more realistic picture of the data (the Adjusted R^2 actually bumped up from 0.726 to 0.759). Ultimately, it proves that we can't just apply a blanket rule for how gender inequality affects national income. The economic penalty for gender inequality is heavily dependent on a nation's baseline level of human development.

References

- [1] Social Science Computing Cooperative (SSCC), University of Wisconsin-Madison. *Regression Diagnostics in R*. Available at: <https://sscc.wisc.edu/sscc/pubs/RegDiag-R/index.html>